

ECE 3317

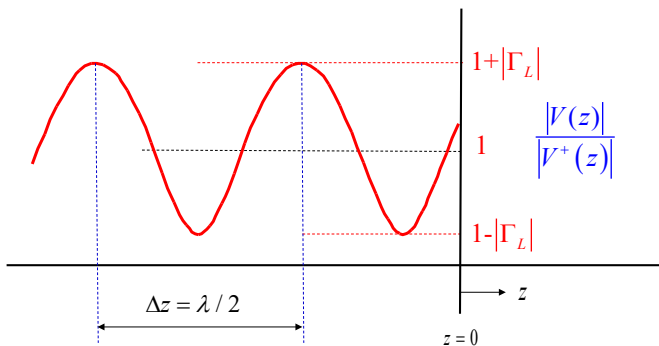
Applied Electromagnetic Waves

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Notes 11

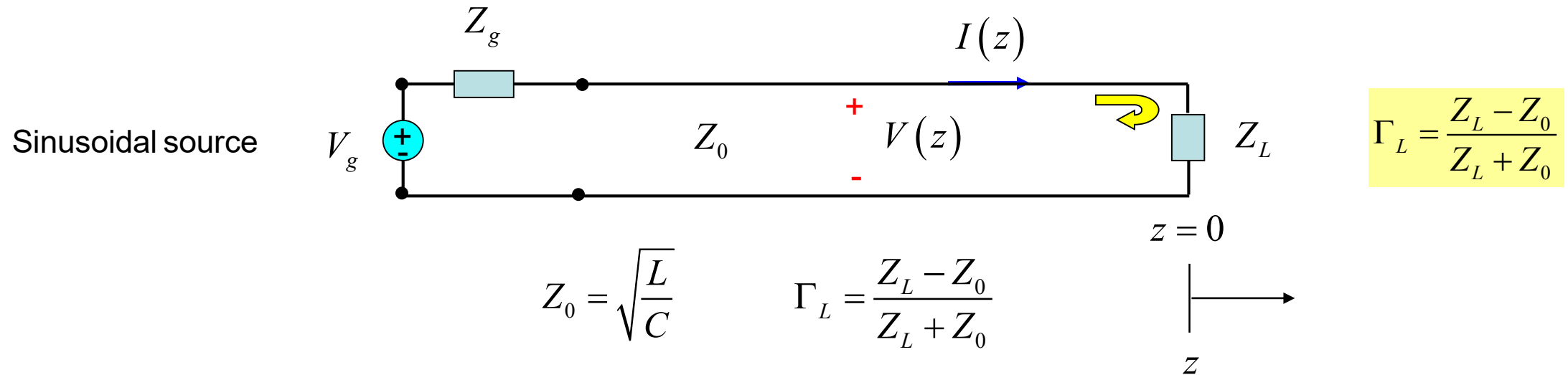
Transmission Lines

(Standing Wave Ratio (SWR) and
Generalized Reflection Coefficient)



Standing Wave Ratio

Consider a lossless transmission line that is terminated with a load:



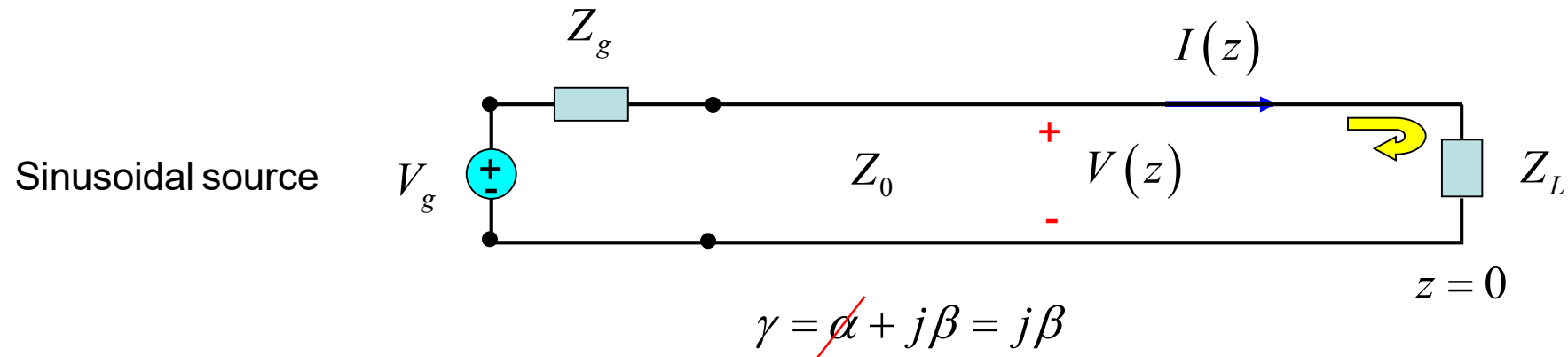
Lossless line:

$$\beta = \omega\sqrt{LC} = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_0\epsilon_0}\sqrt{\mu_r\epsilon_r} = \frac{2\pi f}{c}\sqrt{\mu_r\epsilon_r} = \frac{2\pi}{\lambda_0}\sqrt{\mu_r\epsilon_r} = \frac{2\pi}{\lambda_d}, \quad \lambda_d = \frac{\lambda_0}{\sqrt{\mu_r\epsilon_r}}$$

$$\lambda_0 = \frac{c}{f} \quad c = 2.99792458 \times 10^8 \text{ [m/s]}$$

Standing Wave Ratio (cont.)

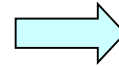
Consider a lossless transmission line that is terminated with a load:



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$V(z) = A(e^{-j\beta z} + \Gamma_L e^{+j\beta z})$$

$$I(z) = \frac{1}{Z_0} A(e^{-j\beta z} - \Gamma_L e^{+j\beta z})$$



$$V(z) = A e^{-j\beta z} (1 + \Gamma_L e^{+j2\beta z})$$

$$I(z) = \frac{1}{Z_0} A e^{-j\beta z} (1 - \Gamma_L e^{+j2\beta z})$$

$$V^+(z) = A e^{-j\beta z}$$

$$V^-(z) = A \Gamma_L e^{+j\beta z}$$

$$V^-(z) / V^+(z) = \Gamma_L e^{+j2\beta z}$$

Standing Wave Ratio (cont.)

$$V(z) = Ae^{-j\beta z} \left(1 + \Gamma_L e^{+j2\beta z} \right)$$

Denote $\Gamma_L = |\Gamma_L| e^{j\phi_L}$ (the polar form of Γ_L)

Then we have: $V(z) = Ae^{-j\beta z} \left(1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right)$

The magnitude is: $|V(z)| = |A| \left| 1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right|$

Maximum voltage: $V_{\max} \equiv |V(z)|_{\max} = |A|(1 + |\Gamma_L|)$ $\phi_L + 2\beta z_{\max}^v = 2\pi m$

Minimum voltage: $V_{\min} \equiv |V(z)|_{\min} = |A|(1 - |\Gamma_L|)$ $\phi_L + 2\beta z_{\min}^v = \pi + 2\pi n$

$$m, n = 0, \pm 1, \pm 2, \dots$$

Standing Wave Ratio (cont.)

The voltage standing wave ratio is the ratio of $V_{\max}$ to $V_{\min}$:

$$\text{VSWR} \equiv \frac{V_{\max}}{V_{\min}}$$

We then have:

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$1 \leq \text{VSWR} \leq \infty$$

↑
Perfect match: $\Gamma_L = 0$

↑
Reactive load: $|\Gamma_L| = 1$

Standing Wave Ratio (cont.)

For the **current** we have:

$$|I(z)| = |A| \left(\frac{1}{Z_0} \right) \left| 1 - |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right|$$

Hence, we have:

$$I_{\max} \equiv |I(z)|_{\max} = |A| \left(\frac{1}{Z_0} \right) (1 + |\Gamma_L|) \quad \phi_L + 2\beta z_{\max}^i = \pi + 2\pi n$$

$$I_{\min} \equiv |I(z)|_{\min} = |A| \left(\frac{1}{Z_0} \right) (1 - |\Gamma_L|) \quad \phi_L + 2\beta z_{\min}^i = 2\pi m$$

Note:

The current is maximum where the voltage is minimum, and vice versa.

The current standing wave ratio is thus:

$$\text{ISWR} \equiv \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

Hence $\text{VSWR} = \text{ISWR} = \text{SWR}$

Standing Wave Pattern

$$|V(z)| = |A| \underbrace{\left| 1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right|}_{\text{Plot this part}}$$

Plot this part

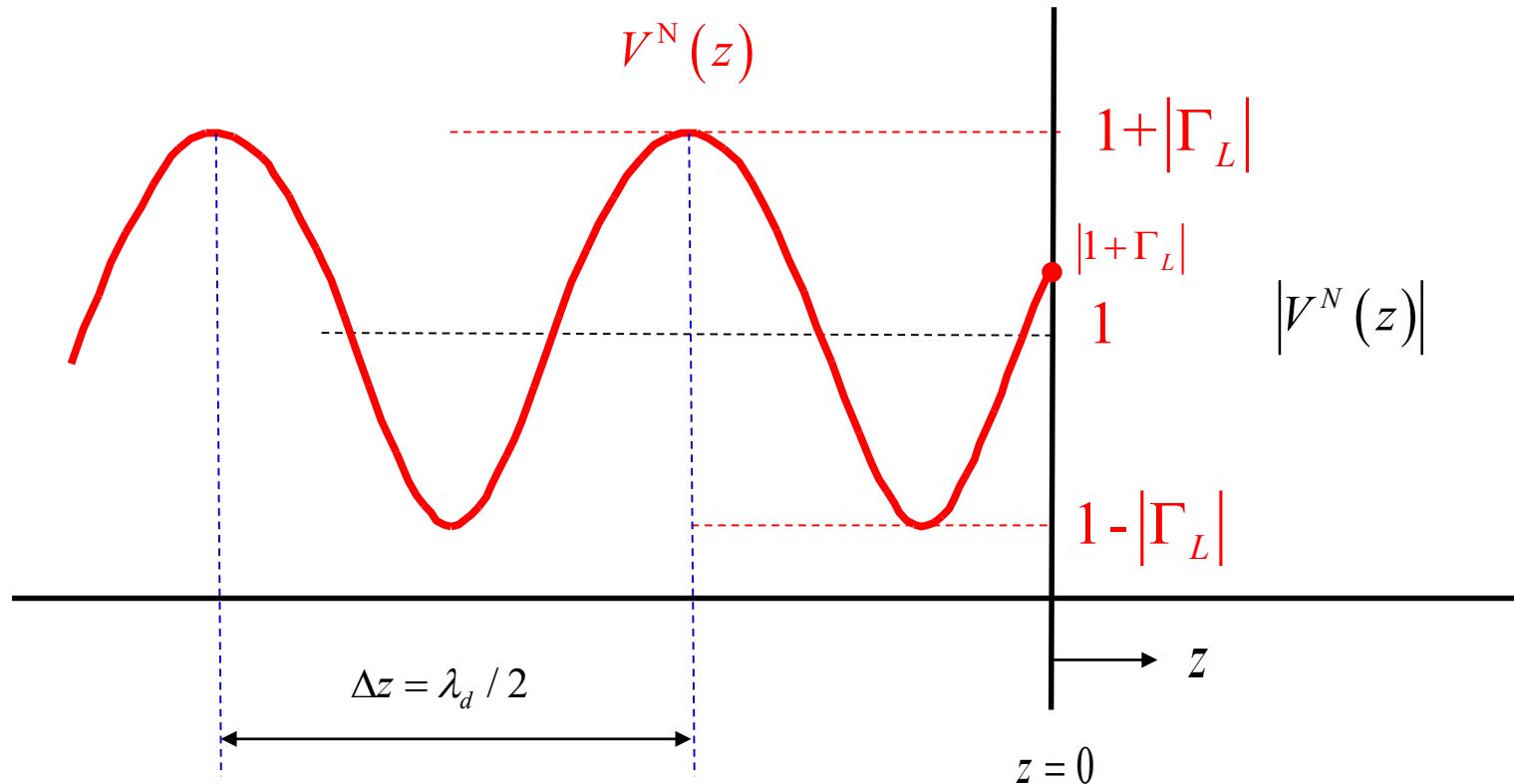
$$|V^N(z)| \equiv \left| \frac{V(z)}{V^+(z)} \right| = \left| 1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right|$$

$$2\beta z = 2 \left(\frac{2\pi}{\lambda_d} \right) z = 4\pi \left(\frac{z}{\lambda_d} \right)$$

Note:

$$V^+(z) = A e^{-j\beta z}$$

$$|V^+(z)| = |A|$$



Standing Wave Pattern (cont.)

For a real load ($Z_L = R_L$):

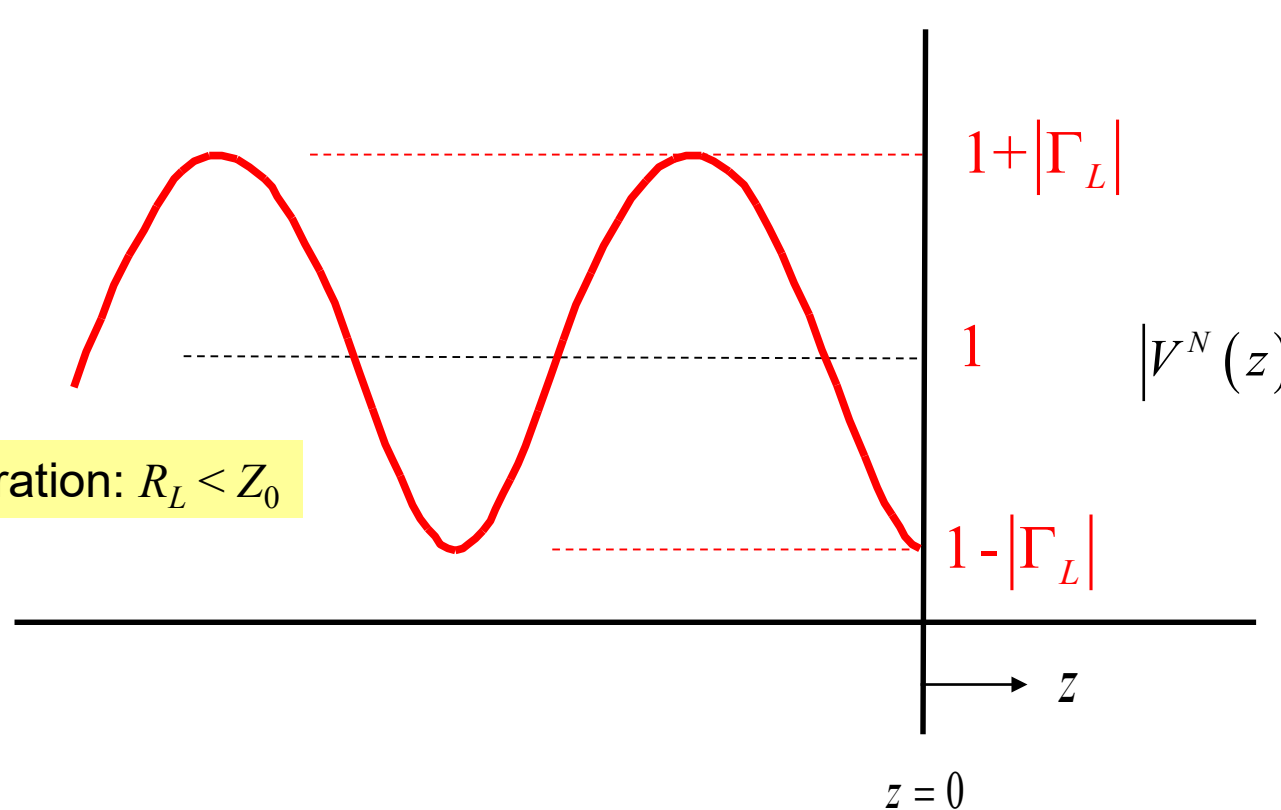
Γ_L is real, and there is always a voltage maximum or minimum at the load ($z = 0$).

$$R_L < Z_0 \Rightarrow \Gamma_L < 0 \Rightarrow \phi_L = \pi \Rightarrow \text{voltage minimum at } z = 0 \quad (\phi_L + 2\beta z_{\min}^v = \pi + 2\pi n)$$

$$R_L > Z_0 \Rightarrow \Gamma_L > 0 \Rightarrow \phi_L = 0 \Rightarrow \text{voltage maximum at } z = 0 \quad (\phi_L + 2\beta z_{\max}^v = 0 + 2\pi m)$$

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = |\Gamma_L| e^{j\phi_L}$$

Illustration: $R_L < Z_0$



The equations in parenthesis are satisfied for $z_{\min, \max} = 0$.

Standing Wave Ratio: Real Load

Special case of a real load impedance $Z_L = R_L$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{R_L - Z_0}{R_L + Z_0}$$

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + \left| \frac{R_L - Z_0}{R_L + Z_0} \right|}{1 - \left| \frac{R_L - Z_0}{R_L + Z_0} \right|}$$

Case a: $R_L \geq Z_0$

$$\text{SWR} = \frac{1 + \left(\frac{R_L - Z_0}{R_L + Z_0} \right)}{1 - \left(\frac{R_L - Z_0}{R_L + Z_0} \right)} = \frac{(R_L + \cancel{Z_0}) + (R_L - \cancel{Z_0})}{(\cancel{R_L} + Z_0) - (\cancel{R_L} - Z_0)}$$

Standing Wave Ratio: Real Load (cont.)

Hence

$$\text{SWR} = \frac{R_L}{Z_0} \quad (R_L \geq Z_0)$$

Case b: $R_L \leq Z_0$

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 - \left(\frac{R_L - Z_0}{R_L + Z_0} \right)}{1 + \left(\frac{R_L - Z_0}{R_L + Z_0} \right)} = \frac{(\cancel{R_L} + Z_0) - (\cancel{R_L} - Z_0)}{(\cancel{R_L} + Z_0) + (\cancel{R_L} - Z_0)}$$

Hence

$$\text{SWR} = \frac{Z_0}{R_L} \quad (R_L \leq Z_0)$$

Standing Wave Ratio: Real Load (cont.)

Hence, for a real load impedance we have:

$$SWR = \max\left(\frac{R_L}{Z_0}, \frac{Z_0}{R_L}\right)$$

(We choose whichever one is greater than 1.)

Example (6.6, Shen and Kong)

Given: $Z_L = 17.4 - j30 \, [\Omega]$ and $Z_0 = 50 \, [\Omega]$

Find: Γ_L , SWR, $z_{\min}^v$, $V_{\max}^N$, $V_{\min}^N$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$



$$\Gamma_L = -0.24 - j0.55$$

$$= 0.6 e^{-j(1.99)}$$

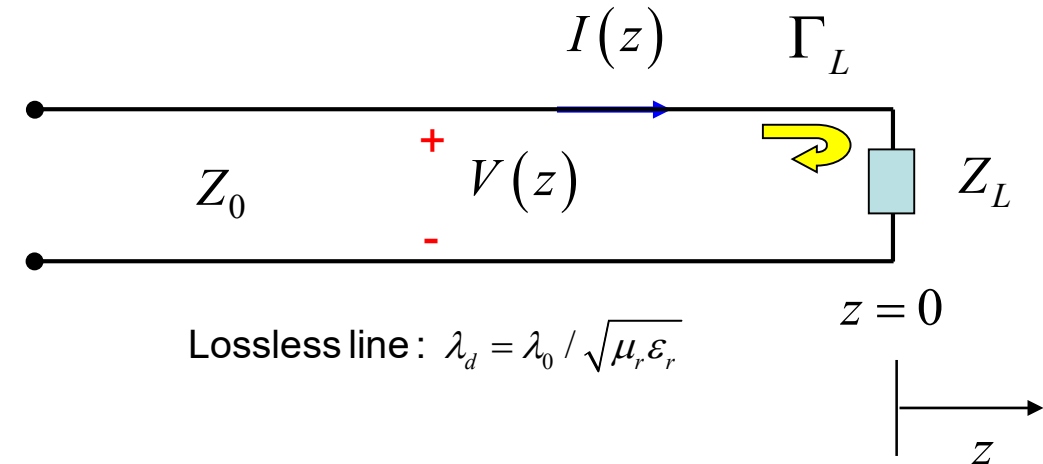
$$|\Gamma_L| = 0.6$$

$$\phi_L = \angle \Gamma_L = -1.99 \, [\text{rad}]$$

$$\text{SWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + 0.6}{1 - 0.6} = 4.0$$

$$V_{\max} \text{ when } \phi_L + 2\beta z_{\max}^v = 0, \pm 2\pi, \dots$$

$$V_{\min} \text{ when } \phi_L + 2\beta z_{\min}^v = \pm \pi, \pm 3\pi, \dots$$



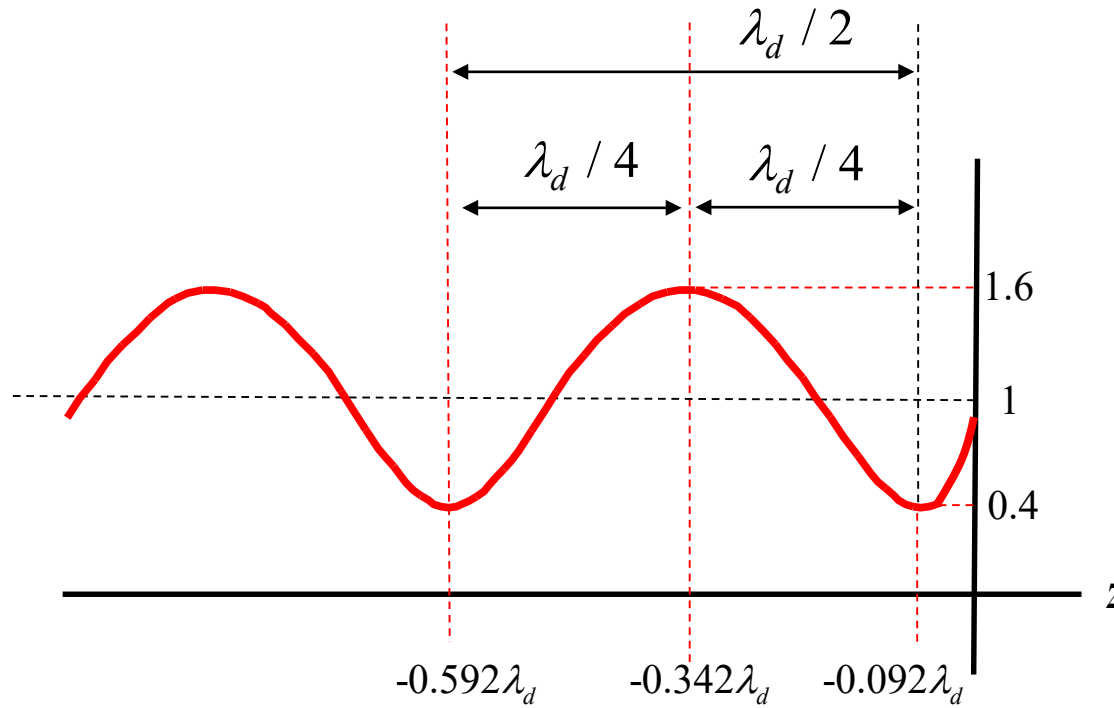
$$V_{\max}^N = V_{\max} / |V^+| = 1 + |\Gamma_L| = 1.6$$

$$V_{\min}^N = V_{\min} / |V^+| = 1 - |\Gamma_L| = 0.4$$

$$z_{\min}^v = \frac{-\pi - \phi_L}{2\beta} = \frac{(-\pi + 1.99)}{2(2\pi / \lambda_d)} = -0.092\lambda_d$$

(Choose $-\pi$: pick the value of $z_{\min}$ closest to the load.)

Example (6.6, Shen and Kong) (cont.)



$$|V^N(z)| = \left| 1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right|$$



$$|V^N(z)| = \left| 1 + (0.6) e^{+j(-1.99 + 4\pi(z/\lambda_d))} \right|$$

$$|V^N(z)|$$

$$\Gamma_L = -0.24 - j0.55$$

$$= 0.6 e^{-j(1.99)}$$

$$\text{SWR} = 4.0$$

$$z_{\min} = -0.092\lambda_d$$

$$V_{\max} / |V^+| = 1.6$$

$$V_{\min} / |V^+| = 0.4$$

Example (6.6, Shen and Kong) (cont.)

Reverse problem:

Given: $\text{SWR} = 4.0$ $z_{\min}^v = -0.092\lambda_d$



This problem has practical significance: often we are interested in figuring out what an unknown load is.

What is the unknown load impedance?

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = 4.0 \quad \Rightarrow \quad \text{SWR}(1 - |\Gamma_L|) = 1 + |\Gamma_L| \quad \Rightarrow \quad |\Gamma_L| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

so

$$|\Gamma_L| = \frac{4.0 - 1}{4.0 + 1} = 0.6$$

$$|\Gamma_L| = 0.6$$

$$z_{\min}^v = \frac{-\pi - \phi_L}{2\beta} = \frac{(-\pi - \phi_L)}{2(2\pi / \cancel{\lambda_d})} = -0.092\cancel{\lambda_d}$$



$$\phi_L = \angle \Gamma_L = -1.99 \text{ [rad]}$$

Solve for ϕ_L .

$$\Gamma_L = 0.6 e^{-j(1.99)}$$

(Note: Any multiple of 2π can be added to ϕ_L .)

Example (6.6, Shen and Kong) (cont.)

$$\Gamma_L = 0.6 e^{-j(1.99)}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} \quad \Rightarrow \quad (Z_L + Z_0)\Gamma_L = Z_L - Z_0 \quad \Rightarrow \quad Z_L = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right)$$

Hence, we have:

$$Z_L = 50 \left(\frac{1 + 0.6 e^{-j(1.99)}}{1 - 0.6 e^{-j(1.99)}} \right)$$

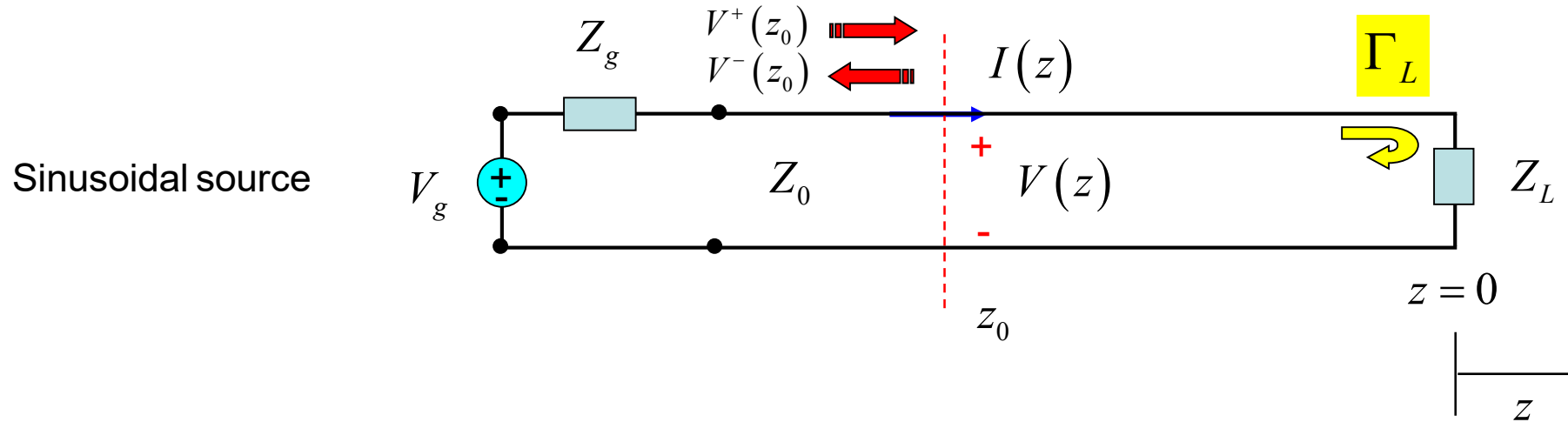
The calculation yields:

$$Z_L = 17.4 - j30 \, [\Omega]$$

We have solved for the unknown load impedance!

Generalized Reflection Coefficient

Define the “generalized reflection coefficient” $\Gamma(z_0)$ at a point z_0 on the line:



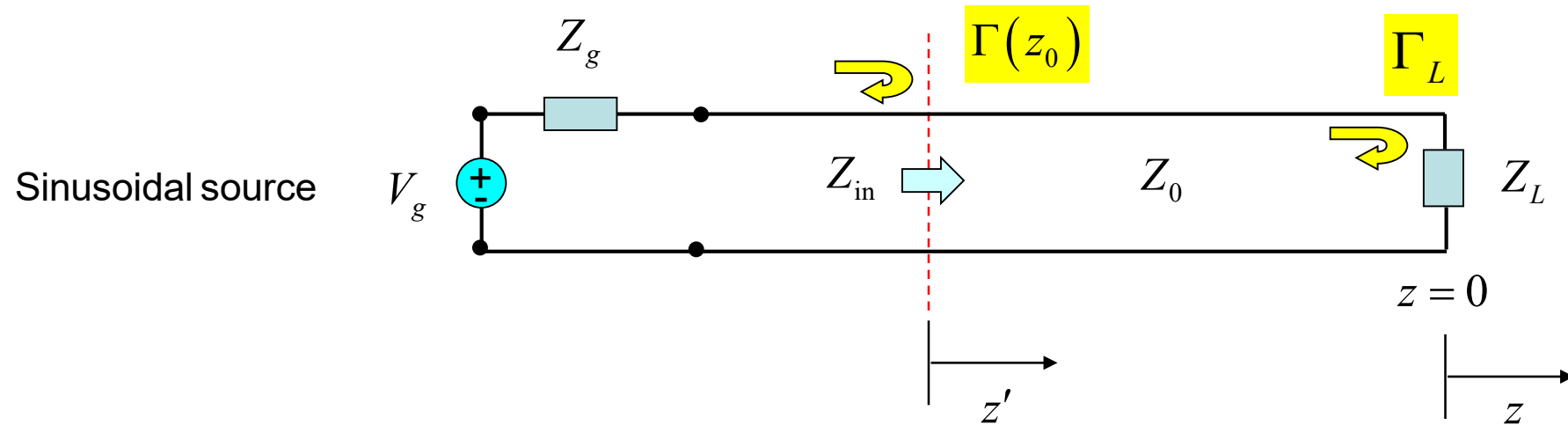
$$V(z) = V^+(z) + V^-(z) = Ae^{-j\beta z} (1 + \Gamma_L e^{+j2\beta z})$$

$$V^+(z) = Ae^{-j\beta z}, \quad V^-(z) = A\Gamma_L e^{+j\beta z}$$

$$\Gamma(z_0) \equiv \frac{V^-(z_0)}{V^+(z_0)} \quad \Rightarrow \quad \Gamma(z_0) = \Gamma_L e^{+j2\beta z_0}$$

Generalized Reflection Coefficient (cont.)

We identify $\Gamma(z_0)$ as the reflection coefficient at the point z_0 , with Z_{in} acting as the load impedance.



Hence

$$\Gamma(z_0) = \frac{Z_{\text{in}}(z_0) - Z_0}{Z_{\text{in}}(z_0) + Z_0}$$

Generalized Reflection Coefficient (cont.)

$$\Gamma(z_0) = \frac{Z_{\text{in}}(z_0) - Z_0}{Z_{\text{in}}(z_0) + Z_0}$$

Hence

$$(Z_{\text{in}}(z_0) + Z_0)\Gamma(z_0) = Z_{\text{in}}(z_0) - Z_0$$



$$Z_{\text{in}}(z_0)[1 - \Gamma(z_0)] = Z_0[1 + \Gamma(z_0)]$$



$$Z_{\text{in}}(z_0) = Z_0 \left(\frac{1 + \Gamma(z_0)}{1 - \Gamma(z_0)} \right)$$

Generalized Reflection Coefficient (cont.)

Define a normalized input impedance at point z_0 :

$$Z_{\text{in}}^{\text{N}}(z_0) \equiv \frac{Z_{\text{in}}(z_0)}{Z_0}$$

We then have, from the last slide:

$$Z_{\text{in}}^{\text{N}}(z_0) = \left(\frac{1 + \Gamma(z_0)}{1 - \Gamma(z_0)} \right)$$

Note:

This can be used as an alternative formula to the “tangent formula” for calculating input impedance at $z_0 = -d$.

$$\Gamma(z_0) = \Gamma_L e^{+j2\beta z_0}$$

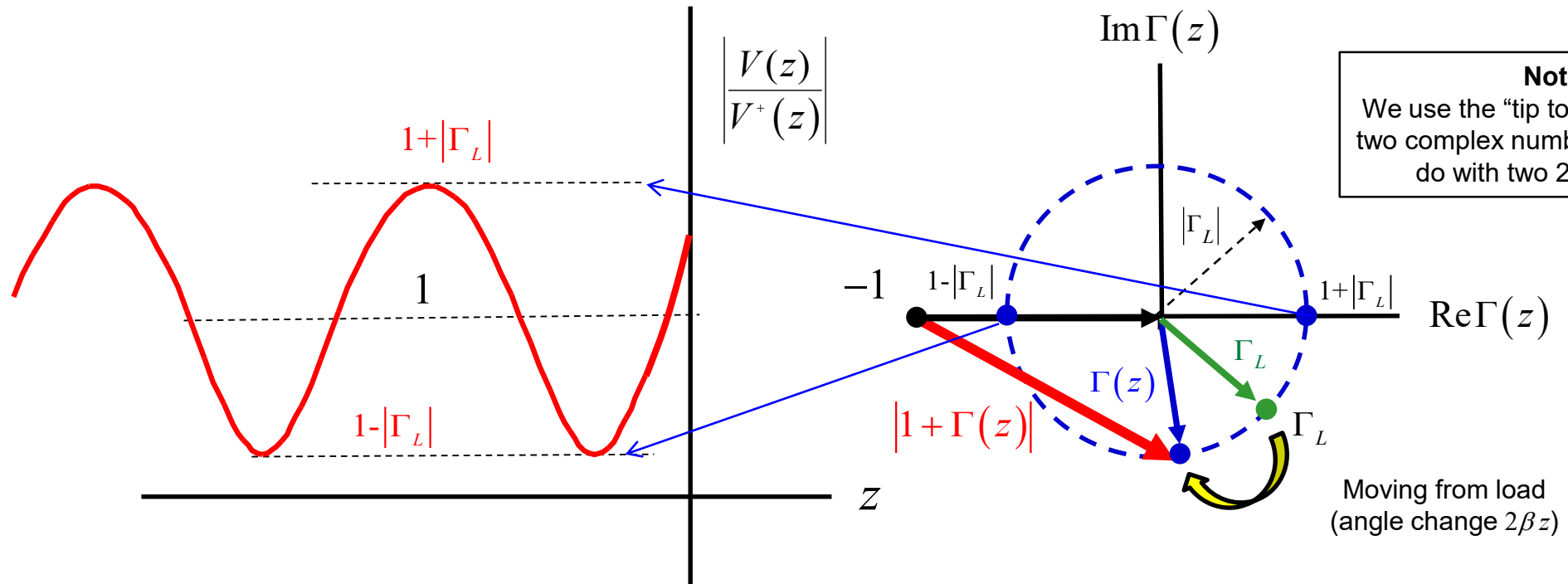
(This is the starting point for the Smith chart discussion.)

Crank Diagram

$$|V^N(z)| = \left| 1 + |\Gamma_L| e^{+j(\phi_L + 2\beta z)} \right| = |1 + \Gamma(z)|$$

where $\Gamma(z) \equiv \Gamma_L e^{+j(2\beta z)}$ $\Gamma_L = |\Gamma_L| e^{+j\phi_L}$

Note: $\Gamma(z) = \frac{V^-(z)}{V^+(z)}$



Note: We go all the way around the crank diagram when $z = -\lambda_d / 2$: $2\beta z = 2 \left(\frac{2\pi}{\lambda_d} \right) \left(-\frac{\lambda_d}{2} \right) = -2\pi$

Example

Given: $Z_L = 100 [\Omega]$ and $Z_0 = 50 [\Omega]$

Calculate the reflection coefficient and the input impedance at $z_0 = -0.125\lambda_d$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{1}{3}$$

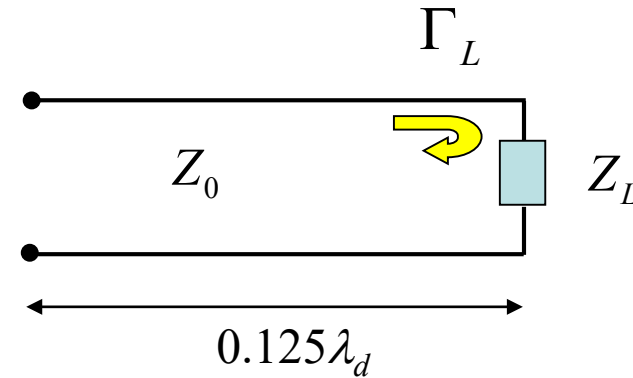
$$\Gamma(z_0) = \Gamma_L e^{j2\beta z_0}$$

$$= \left(\frac{1}{3}\right) e^{j2\left(\frac{2\pi}{\lambda_d}\right)(-0.125\lambda_d)} = \left(\frac{1}{3}\right) e^{-j\pi/2}$$

so

$$\Gamma(z_0) = -j/3$$

$$Z_{\text{in}}^N(z_0) = \left(\frac{1 + \Gamma(z_0)}{1 - \Gamma(z_0)}\right) = \frac{1 - j/3}{1 + j/3} = 0.8 + j(0.6)$$



Lossless line: $\lambda_d = \lambda_0 / \sqrt{\mu_r \epsilon_r}$

$$Z_{\text{in}}(z_0) = (50) Z_{\text{in}}^N(z_0)$$

so

$$Z_{\text{in}}(z_0) = 40 + j(30) [\Omega]$$

Appendix: Summary of Formulas

$$\text{SWR} \equiv \frac{V_{\max}}{V_{\min}}$$

$$\text{SWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\text{SWR} = \max\left(\frac{R_L}{Z_0}, \frac{Z_0}{R_L}\right)$$

$$|\Gamma_L| = \frac{\text{SWR} - 1}{\text{SWR} + 1}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma(z_0) \equiv \frac{V^-(z_0)}{V^+(z_0)}$$

$$\Gamma(z_0) = \Gamma_L e^{+j2\beta z_0}$$

$$\Gamma(z_0) = \frac{Z_{\text{in}}(z_0) - Z_0}{Z_{\text{in}}(z_0) + Z_0}$$

$$Z_{\text{in}}(z_0) = Z_0 \left(\frac{1 + \Gamma(z_0)}{1 - \Gamma(z_0)} \right) \quad Z_L = Z_0 \left(\frac{1 + \Gamma_L}{1 - \Gamma_L} \right) \quad Z_{\text{in}}^{\text{N}}(z_0) \equiv \frac{Z_{\text{in}}(z_0)}{Z_0}$$